

Robust Federated Learning Against Adversarial Attacks

Securing Collaborative Medical AI from Malicious Participants

Team : Transformers

Week 1 — Project Scope

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1. Introduction

Federated Learning (FL) enables multiple clients to collaboratively train machine learning models while keeping data locally stored, making it particularly suitable for healthcare applications where privacy is essential. However, FL systems remain vulnerable to adversarial or Byzantine attacks, where malicious clients intentionally manipulate local updates to corrupt the global model. Such attacks can significantly reduce model reliability and negatively affect medical decision-making. This project aims to study adversarial attacks in Federated Learning and evaluate robust aggregation strategies capable of securing collaborative medical AI systems against malicious participants.

2. Problem Statement

Traditional Federated Learning algorithms such as Federated Averaging (FedAvg) assume that all participating clients behave honestly. However, this assumption is unrealistic in practical distributed environments.

A malicious client may intentionally attempt to poison the learning process through:

- **Data Poisoning:** manipulating local training samples,
- **Model Poisoning:** sending corrupted gradients or parameters,
- **Byzantine Attacks:** generating adversarial updates to destabilize convergence.

Even a small number of malicious participants can severely degrade the global model and compromise prediction reliability.

The main objective of this project is therefore to answer the following question:

How can Federated Learning systems remain robust and reliable in the presence of malicious clients?

To investigate this problem, the project will simulate adversarial participants and compare standard aggregation methods with Byzantine-resilient aggregation strategies.

3. Methodology and Objectives

The project will use a lightweight subset of the **CheXpert** dataset developed by Stanford University for chest X-ray classification tasks.

Dataset Link:

<https://stanfordmlgroup.github.io/competitions/chexpert/>

The proposed methodology consists of simulating a Federated Learning environment involving multiple distributed clients. While most clients will behave honestly, some participants will intentionally act maliciously by poisoning local updates before transmission to the central server.

The project objectives are summarized below:

1. Build a baseline Federated Learning pipeline using Federated Averaging (FedAvg).
2. Simulate adversarial attacks including:
 - Data poisoning,
 - Model poisoning.
3. Implement robust aggregation defenses such as:
 - Krum,
 - Coordinate-wise Median,
 - Trimmed Mean.
4. Compare naive and robust aggregation methods under adversarial settings.
5. Evaluate system robustness using:
 - Accuracy degradation,
 - Attack Success Rate (ASR),
 - Aggregation computation time.

The standard Federated Averaging mechanism is defined as:

$$w_{t+1} = \sum_{k=1}^K \frac{n_k}{n} w_k$$

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Although computationally efficient, this aggregation strategy is highly vulnerable because malicious updates directly influence the global model.

4. Initial State-of-the-Art (SOTA)

Recent studies in secure Federated Learning demonstrate that Byzantine attacks can significantly impact distributed AI systems even when only a small fraction of clients are malicious.

To address this issue, several robust aggregation techniques have been proposed.

Krum

Krum selects the update that is closest to the majority of client updates, reducing the influence of outliers and adversarial gradients.

Coordinate-wise Median

This method computes the median value for each parameter dimension independently instead of averaging all updates. Median operations are naturally more resistant to extreme malicious values.

Trimmed Mean

Trimmed Mean removes suspiciously large and small parameter updates before averaging the remaining values, limiting the effect of poisoned gradients.

The project will experimentally compare:

FedAvg under attack
vs
Robust aggregation under attack
vs
Unattacked FedAvg

to evaluate the effectiveness of Byzantine-resilient Federated Learning strategies in collaborative medical AI systems.

"Privacy-preserving AI must also be secure and trustworthy."